

xDNN: Inference for Deep Convolutional Neural Networks

PAOLO D'ALBERTO, VICTOR WU, AARON NG, RAHUL NIMAIYAR, and
ELLIOTT DELAYE, [Xilinx](#)
ASHISH SIRASAO, [FaceBook](#)

We present xDNN, an end-to-end system for deep-learning inference based on a family of specialized hardware processors synthesized on Field-Programmable Gate Array (FPGAs) and Convolution Neural Networks (CNN). We present a design optimized for low latency, high throughput, and high compute efficiency with no batching. The design is scalable and a parametric function of the number of multiply-accumulate units, on-chip memory hierarchy, and numerical precision. The design can produce a scale-down processor for embedded devices, replicated to produce more cores for larger devices, or resized to optimize efficiency. On [Xilinx](#) Virtex Ultrascale+ VU13P FPGA, we achieve 800 MHz that is close to the Digital Signal Processing maximum frequency and above 80% efficiency of on-chip compute resources.

On top of our processor family, we present a runtime system enabling the execution of different networks for different input sizes (i.e., from 224×224 to 2048×1024). We present a compiler that reads CNNs from native frameworks (i.e., MXNet, Caffe, Keras, and Tensorflow), optimizes them, generates codes, and provides performance estimates. The compiler combines quantization information from the native environment and optimizations to feed the runtime with code as efficient as any hardware expert could write. We present tools partitioning a CNN into subgraphs for the division of work to CPU cores and FPGAs. Notice that the software will not change when or if the FPGA design becomes an ASIC, making our work vertical and not just a proof-of-concept FPGA project.

We show experimental results for accuracy, latency, and power for several networks: In summary, we can achieve up to 4 times higher throughput, 3 times better power efficiency than the GPUs, and up to 20 times higher throughput than the latest CPUs. To our knowledge, we provide solutions faster than any previous FPGA-based solutions and comparable to any other top-of-the-shelves solutions.

CCS Concepts: • **Hardware** → **Power estimation and optimization**; • **Computing methodologies** → **Computer vision**; • **Computer systems organization** → **Systolic arrays**; • **Software and its engineering** → **Software prototyping**;

Additional Key Words and Phrases: AI inference, low latency, high efficiency, custom architectures, and optimizations

ACM Reference format:

Paolo D'Alberto, Victor Wu, Aaron Ng, Rahul Nimaiyar, Elliott Delaye, and Ashish Sirasao. 2022. xDNN: Inference for Deep Convolutional Neural Networks. *ACM Trans. Reconfigurable Technol. Syst.* 15, 2, Article 18 (January 2022), 29 pages.

<https://doi.org/10.1145/3473334>

Authors' addresses: P. D'Alberto, V. Wu, A. Ng, R. Nimaiyar, and E. Delaye, [Xilinx](#), Logic Drive, San Jose, CA 95124; emails: {paolod, viw, aaronn, rahuln, elliott}@xilinx.com; A. Sirasao, [FaceBook](#), 1 Hacker Way, Menlo Park, CA 94025; email: asirasao@fb.com.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.

1936-7406/2022/01-ART18 \$15.00

<https://doi.org/10.1145/3473334>

ACM Transactions on Reconfigurable Technology and Systems, Vol. 15, No. 2, Article 18. Pub. date: January 2022.